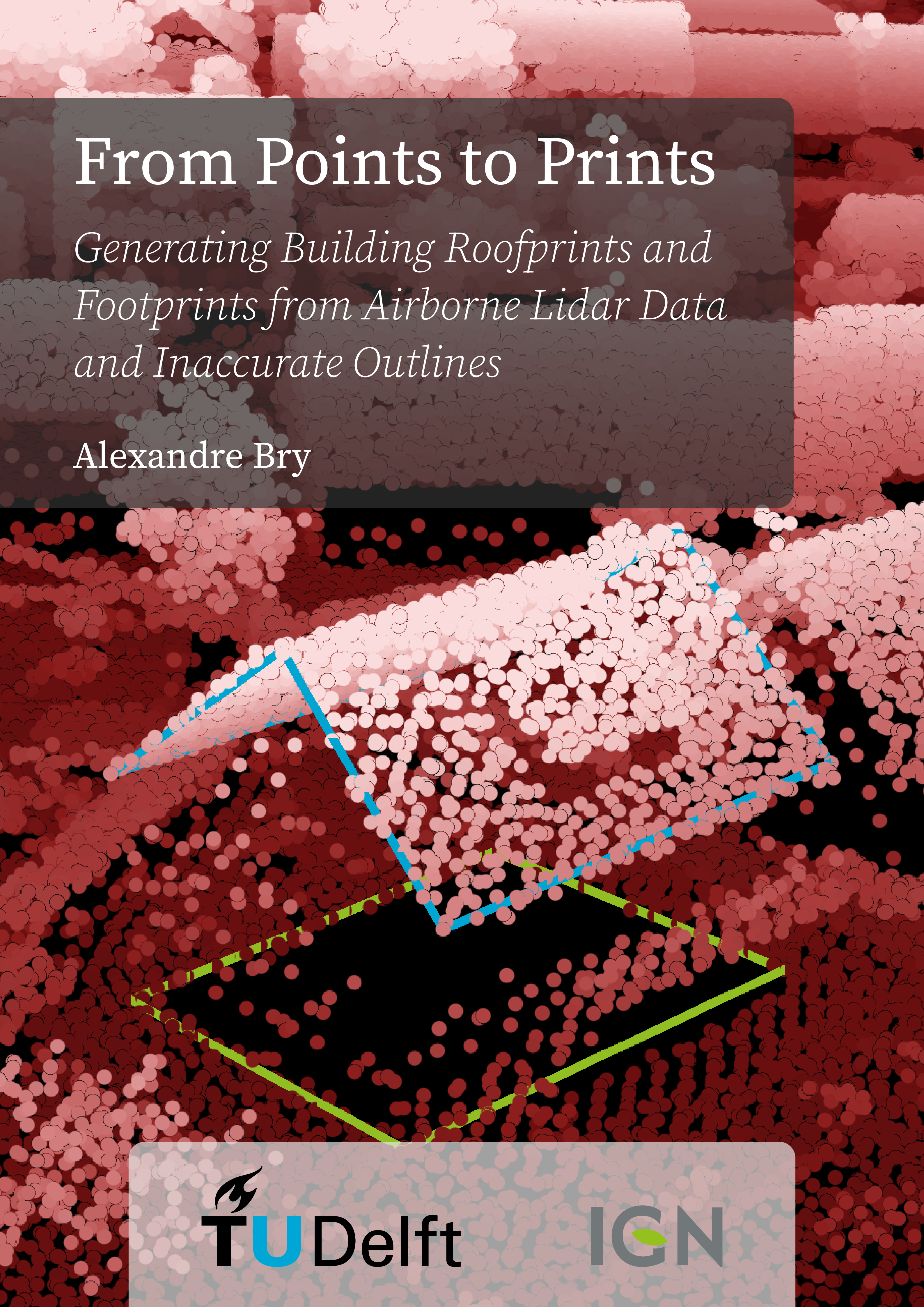

From Points to Prints

Generating Building Roofprints and Footprints from Airborne Lidar Data and Inaccurate Outlines

Alexandre Bry

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MSc Thesis Report

by

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Abstract

In recent years, several methods have been successfully developed to create 3D building models in LoD2.2 from ALS point clouds. These methods however rely on 2D horizontal building outlines whose definitions and quality vary significantly. There are mainly two types of outlines: roofprints and footprints, and having access to both of them allows to reconstruct the façades and the roof independently, therefore improving from LoD2.2 to LoD2.3 models. Moreover, this distinction also allows more precise analyses depending on the use case. Therefore, we present a method to construct both a roofprint and a footprint from ALS data and existing inaccurate outlines. By first precisely deforming and aligning the outline on the roof to turn it into the roofprint, our method can then accurately identify the few ground and façade points available, if there are any, in order to produce a footprint coherent with the roofprint. Our method preserves the validity of the polygons and the topological relations between them. It has been tested successfully on the French national datasets BD TOPO and LiDAR HD.

Acknowledgements

First, I want to thank my two main supervisors Hugo Ledoux and Bruno Vallet for their steady support over the whole course of the thesis, and for their very valuable ideas and suggestions. The many discussions we had were crucial to take pragmatic and thoughtful decisions all along, and to keep my motivation high.

I want to extend my acknowledgements to everyone who followed my thesis for the regular monthly meetings and came with interesting and refreshing ideas with their viewpoints from the outside. This includes Ravi Peters as my second supervisor for TU Delft, but also Emmanuel Séguin, Teng Wu, Florent Geniet and François Becirspahic on the side of the IGN.

Then, I have to thank all my colleagues from IGN especially in the LASTIG and in the SIMV department who welcomed me during the 5 months I spent there. Because in an intense and self-centred work like research, taking some time for breaks and to enjoy other peoples company is necessary to keep the mental and the ideas flourishing.

Finally, I am really grateful to Élodie, to my family and to my friends, who always supported me whether mentally or financially. They enabled me and encouraged me to live this one-and-a-half year experience in the Netherlands, and welcomed me back in France as if I had never left.

Outline

1. Introduction

In 2024, the Institut national de l'information géographique et forestière (IGN)[1] and two other French public entities have launched an initiative to bring together partners who can contribute to the development of a nation-wide digital twin (IGN, 2024a). This initiative was officially launched in April 2026 under the name of JUNN. The idea behind a digital twin is to gather in one interface many sources and types of data, in order to make them more discoverable, accessible and usable, and by doing so allow for simulations and discussions based on real, accurate and complex data. Ecological planning and sustainable land use are presented as some of the priorities this project should accommodate. In a different post, it is explained how the LiDAR HD[2] — the first project to collect high-density point clouds on almost the whole territory of France — is central to the future digital twin (IGN, 2024b). This comes from the unprecedented precision that it brings compared to previous data used and maintained by the IGN.

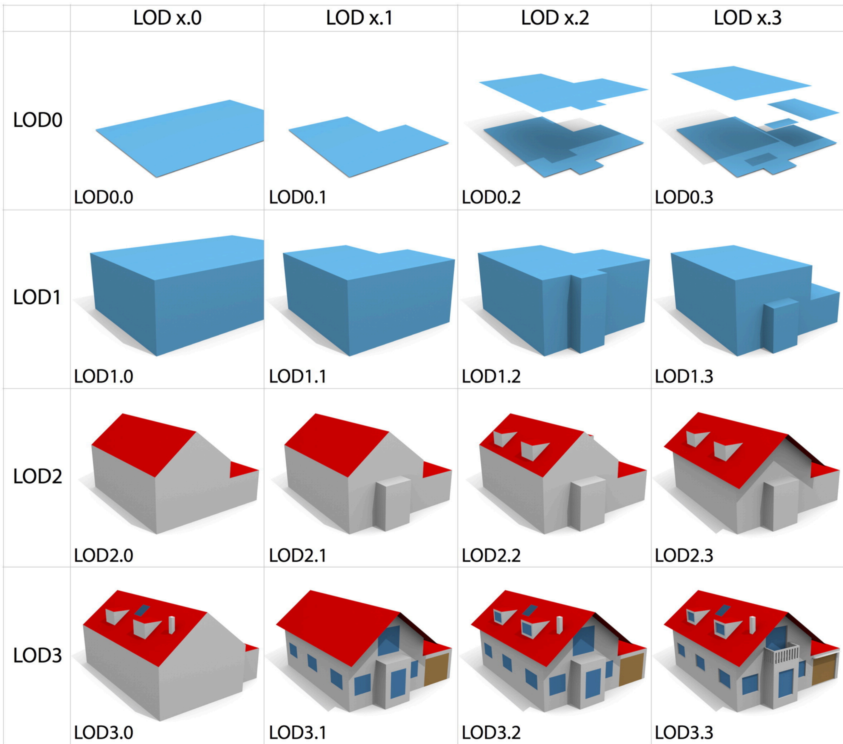


Figure 1: Visual example of the refined LoDs for a residential building (Biljecki et al., 2016).

One of the many components of the future digital twin is buildings. Many algorithms have been developed to try to reconstruct structured and accurate 3D building models from various data sources, including point clouds. To characterise the properties of the buildings created by these different methods, 5 Levels of Detail (LoDs) have been introduced by Gröger et al. (2012), before being extended into 16 LoDs described by Biljecki et al. (2016) and are illustrated in Figure 1. One of the current state-of-the-art algorithms was created by researchers at Delft University of Technology (TU Delft) and is called roofer[3] (Pađen et al., 2024). It was applied to the whole of the Netherlands to create the 3DBAG, the first complete dataset of Dutch buildings in LoD2.2 (Peters et al., 2022). This algorithm however requires two input data: a dense Airborne Laser Scanning (ALS) 3D point cloud and 2D building outlines. In the Netherlands, the Actueel Hoogtebestand Nederland (AHN) was used as the point cloud and the Basisregistratie Adressen en Gebouwen (BAG) was used as the outlines.

[1] https://www.ign.fr/institut/identity-card
[2] https://www.data.gouv.fr/datasets/nuages-de-points-lidar-hd
[3] https://github.com/3DBAG/roofer

However, talking about a building outline is imprecise, as there are many different ways to create a 2D horizontal representation of a building. There are mainly two kinds of 2D building outlines that we consider in this thesis:

- the footprint, defined as the horizontal 2D polygon obtained by projecting vertically the *outer walls/façades* of a building,
- the roofprint, defined as the horizontal 2D polygon obtained by projecting vertically the *roof* of a building and taking its outer boundary.

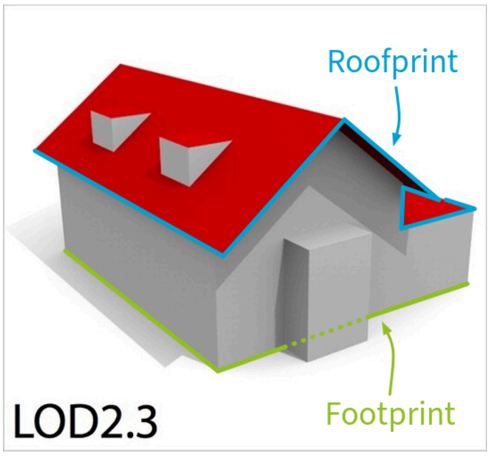


Figure 2: Visual definition of the roofprint and footprint on the LoD2.3 building example by Biljecki et al. (2016). The roofprint in this picture displays the edges of the roof used for the definition and needs to be projected vertically to get the actual roofprint.

These definitions are illustrated in Figure 2. In the rest of this document, I will use the terms roofprint and footprint when possible, and otherwise use outline for a more general term. Usually, due to roof overhangs and gutters, the roof extends further than the walls, meaning that the footprint is included in the roofprint. As an example, the roofprint is what matters in estimating solar energy potential — in combination with other factors such as roof orientation and angle. But in many other applications — such as taxes or energy consumption — an accurate estimation of the area and volume of buildings is necessary, which will be better with a footprint.

The IGN already has a dataset containing building outlines, called BD TOPO. However, this dataset has some issues, that can be explained by how it was historically built from different sources. Most outlines (88.54%) come from terrain measurements and are therefore footprints, but most of the rest (10.09%) come from aerial image detection and are therefore roofprints. A small proportion was automatically generated from the LiDAR HD (0.36%) and the origin of the rest is not specified (1.01%)[4]. The georeferencing of these building outlines is often wrong by up to a few meters. This makes combining them with correctly georeferenced point clouds more complicated.

All in all, the current context combines:

- an objective to build a digital twin of France, including 3D buildings with algorithms which would benefit from or require correct building outlines (such as roofer),
- newly available data with high precision and correct georeferencing (LiDAR HD),
- the example of the Netherlands where a great dataset of 3D building models was built from similar point cloud data (3DBAG from AHN),
- an interesting and not yet fully explored question of the possibility of extracting both an accurate footprint and an accurate roofprint from point clouds,
- an existing dataset that provides nation-wide and potentially great data but is however missing harmonisation and precise georeferencing (BD TOPO).

[4]These numbers come from the 2026-03-15 version of the BD TOPO.

Therefore, the research question of this thesis is:

> ***How to generate coherent building roofprints and footprints from high-density ALS point clouds and existing imprecise outlines?***

A few more specific sub-questions are also addressed in this thesis:
- How to identify and use the points on roof edges in ALS point clouds?
- How to identify and use the points in an ALS point cloud that contain information about the façades despite their sparsity?
- How to deform an imprecise outline with global and local transformations while preserving the angles of the edges?

The core of this report is structured as a research paper, targeted at an expert audience, which can be found in *Chapter 3*. To make its content accessible to non-experts, preliminary materials on several different topics are provided in *Chapter 2*, although they still assume some GIS-related knowledge, to the level of the TU Delft Master of Geomatics. Finally, the report ends with a conclusion and a presentation of potential future work on the topic.

2. Preliminary Materials

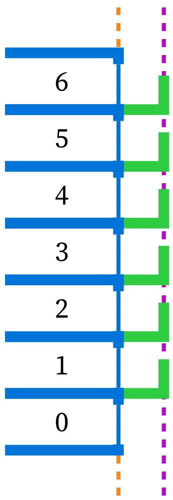
This chapter aims at introducing the knowledge and vocabulary necessary to understanding the core of the thesis (*Chapter 3*). It is intended for readers who are not expert in the topic at hand, however the *Chapter 3* is self-contained.

2.a. Roofprints and footprints

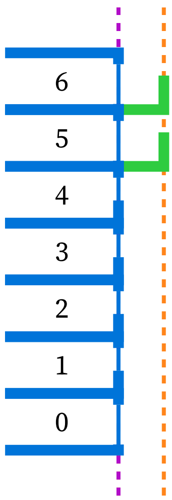
As presented in *Chapter 1*, the difference between roofprints and footprints is central for this thesis. One of the reasons why both of them are important is that different sources of data often make it easier to get one of the two:

- surveyors in the field mostly use the walls and therefore measure the footprint,
- experts working on aerial imagery can only use the roof as some walls will not be visible, meaning that they measure the roofprint,
- ALS point clouds (such as the LiDAR HD and the AHN) give many points on the roofs and therefore make it easier to extract the roofprint,
- Terrestrial Laser Scanning (TLS) and Mobile Laser Scanning (MLS) point clouds give many points on the walls and therefore make it easier to extract the footprint.

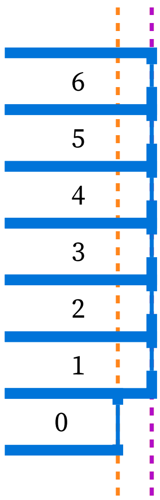
Even with the definitions given before, some aspects still need to be clarified to be able to unequivocally draw a roofprint and a footprint for every building. First, the inclusion of balconies and other outdoor elements of buildings is unclear in many scenarios, as illustrated in Figure 3. In blocks of flats with multiple storeys where every storey except the ground floor has the same exact balcony (see Figure 3a), does the end of the balcony become the position of the façade? Sometimes, the whole building is extruded horizontally except the ground floor (see Figure 3c), in a structure which looks like balconies except they are not open, are these considered as balconies? There are many more questions to consider when trying to properly characterise 2D outlines to create a coherent dataset, and many of these questions do not have a right or wrong answer: the best answer depends on the final application. There are even cases, such as buildings with non-vertical façades, which completely break the purpose of defining a simple and unique 2D outline.



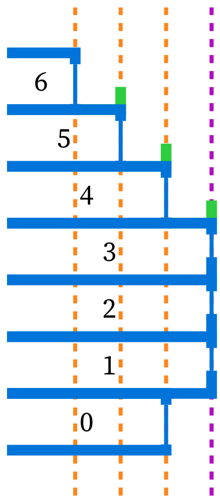
(a) Same balcony at every storey.



(b) Only a few balconies.



(c) Extruded façade.



(d) Different façades with some balconies.

Figure 3: Illustration in profile view of the unclear definition of the façade with balconies. The buildings are in blue with thin lines representing windows and the balconies in green. Dotted lines represent potential façades and the purple one is the one we chose in our definition.

In our case, the potential usage of footprints and roofprints which is considered is the possibility to turn LoD2.2 buildings into LoD2.3 by modelling roof overhangs, as shown in Figure 1 (*Chapter*

1). Since methods like roofer use the points from the roof to reconstruct buildings, they require a roofprint to work properly. But then, since there is only a sparse distribution of points on the façades (if there are any points), identifying planes is often impossible, and the roofs are simply extruded down, creating LoD2.2 buildings. If the footprint was known precisely, it would be possible to instead extrude it up to the roof, assuming that it is contained in the roofprint.

Therefore, our definitions for footprints and roofprints should be the ones that would lead to the most accurate reconstruction of the buildings in 3D. In practice, this means that the façades are defined as the most prevalent vertical surfaces, or in other terms, the vertical plane which has the largest intersection with the actual building (see Figure 3). This will often be the end of the balcony when the exact same balcony is present at every storey such as in Figure 3a. With this definition, the balconies have a significant advantage because in ALS data, a balcony occludes the façade below it in the same way as roof overhangs, so they prevent the potential façade behind them to be seen.

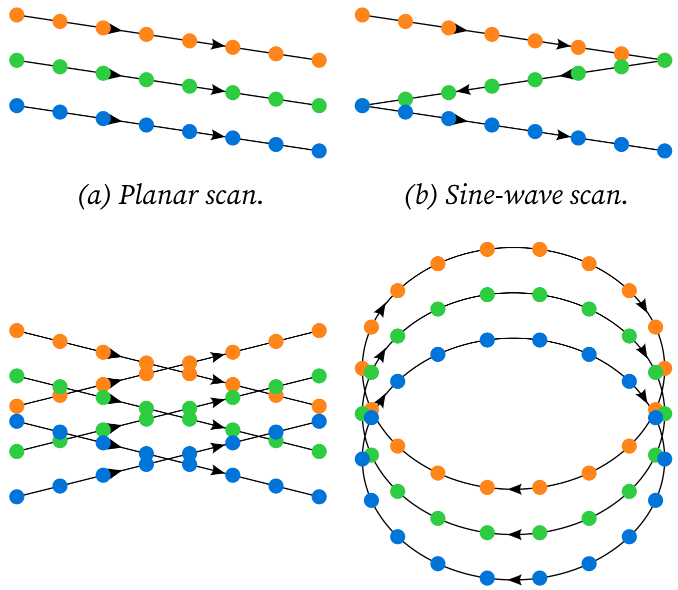
2.b. ALS point clouds

Airborne Laser Scanning (ALS) point clouds are sets of points acquired by a Light Detection and Ranging (lidar) sensor mounted on a flying vehicle. These lidar sensors emit laser pulses at a very high frequency with a varying direction. At the same time, the vehicle usually moves as much as possible in a straight line, at a constant speed and constant height. Each pulse results in a continuous return signal, where peaks correspond to objects on the pulse trajectory, and are called echos. By combining the trajectory of the scanning vehicle, the angle of the lidar sensor and the time between emitting the pulse and receiving the returned peak, a 3D position can be computed for each echo, corresponding to an object in the scene.

The characteristics of the point clouds obtained with this method depend on many parameters (Wu et al., 2026):

- the pulse frequency: a higher pulse frequency results in denser point clouds along the scan lines,
- the altitude of the vehicle: a lower altitude results in denser point clouds covering smaller areas,
- the speed of the vehicle: a slower vehicle results in denser point clouds along the direction of the vehicle,
- the type of lidar sensor: one or multiple fibres (laser emitters) with different motions (planar rotations, conical rotations, oscillations, or even more complex motions) result in different structures.

During this thesis, the main focus was the LiDAR HD, which present similar characteristics to its Dutch counterpart the AHN. However, multiple different ALS sensors have been used by the different contractors who took part in the acquisition of the LiDAR HD. Based on the work of Wu et al. (2026), the different motions for the sensors found in the LiDAR HD are illustrated in Figure 4.



(a) Planar scan. (b) Sine-wave scan.

(c) Cross scan. (d) Circular scan.

Figure 4: Illustration of different types of ALS sensors, with multiple consecutive scan lines in orange, green and blue (the scanning vehicle is moving towards the bottom of the figures), inspired from (Wu et al., 2026).

Even though they look completely different, these share common characteristics. First, the motions of the fibres are continuous except for potential jumps between the end of a scan line and the beginning of the next scan line. This means that except for rare exceptions, two consecutive pulses are very close geometrically. Then, the directions of the pulses are almost vertical, with an angle rarely reaching more than 20°. This property will be crucial for our method, as will be explained later. Finally, we can always reconstruct a sort of multi-dimensional topology over the point cloud, because:

- all the echos acquired from a single pulse are ordered,
- all the pulses acquired along a single scan line are ordered,
- all the scan lines acquired along a single flight strip are ordered.

Therefore, one can move along any of these three dimensions to make computations in a comparable and predictable way. Compared to the inherently chaotic nature of 3D point clouds, with every point having potentially a very different geometric neighbourhood, this can facilitate studying the properties of the point cloud. More specifically, since the flight strips often overlap, they are merged into a point cloud which topological structure can be defined as illustrated in Figure 5. In the LiDAR HD, a tile corresponds to a square area of 1 km by 1 km.

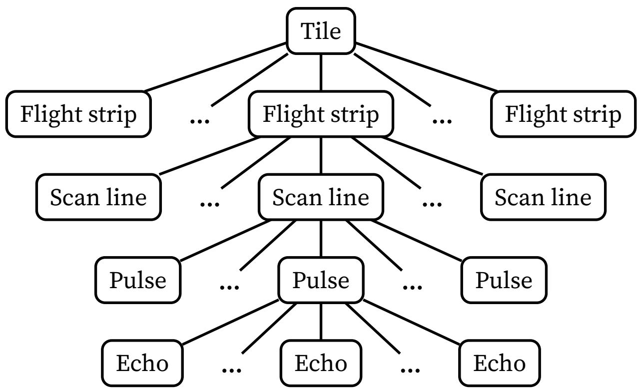


Figure 5: Illustration of the topological structure of a tile of an ALS point cloud.

2.c. Polygon deformation and topology

Polygons are complex shapes on which many different operations are possible. A significant problem about them is the complexity associated with verifying their validity. For example, a simple and brute-force approach to verifying that no pair of edges intersect in a polygon requires $n(n-1)$ checks if the polygon has n edges.

In that regard, modifying a polygon instead of creating a new one from scratch can have some good properties. Depending on the operations that are performed, checking the validity of the polygon can be very simple, or even unnecessary. This is the case for most of the simple global operations on polygons: translations, rotations, scaling and even skewing, as long as they are applied globally, will never break the validity of a polygon. This is because these are rigid transformations: the relative positions of the vertices remain unchanged. However, these operations are very limited in their capabilities: their global aspect prevents them from patching specific regions of the polygons in different ways.

This is where local operations are necessary. For these operations, the two most basic elements to work with are vertices and edges. Choosing between the two is crucial and completely depends on the applications. This choice and the number of constraints enforced in the movement determine how much freedom and how much complexity will be associated with the operation. Here are a few examples, among many different possibilities :

- translating a point freely gives 2 dimensions of freedom: one dimension for each axis (x and y).
- translating the line associated with an edge along its normal gives only 1 dimension of freedom: the distance between the initial and the final line.
- translating and rotating the line associated with an edge gives 2 dimensions of freedom: one for the translation distance and one for the rotation angle.

For most of these local operations, there will be bounds to how much each element can be modified before breaking the validity of the polygon. These bounds correspond to a simple interval if there is only 1 dimension of freedom, and become more and more complex as the number of dimensions of freedom increases. On top of that, these bounds are not independent for every point or edge. Moving one point has an influence on how far its neighbour can be moved at the same time without breaking the polygon. Therefore, the complexity of these operations increases when performing many of them at the same time, and depending on the type of operation, it can be very complex to express and manipulate these relations.

For this thesis, the constraints that we imposed on the initial polygons were the following (more details in *Chapter 3*):

- do not rotate any edge,
- do not flip any edge (see Figure 6).

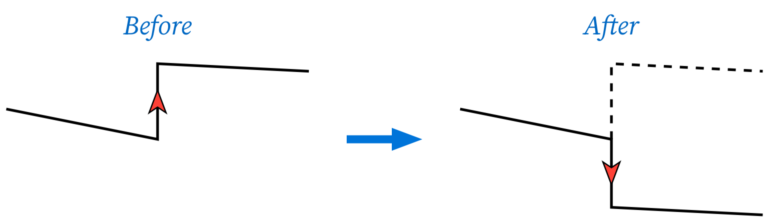


Figure 6: Illustration an edge being flipped by the translation of a neighbour edge.

With these constraints in mind, the simplest way to understand and express the remaining movements is the following. First, each edge is replaced by the infinite line that passes through it. The polygon is therefore defined by a set of ordered lines, which intersections define the vertices (i.e. the two ends of the edges). The only modification allowed is to replace any line by any parallel line, as long as this does not break the validity of the polygon. In practice, this means that there can be a maximum distance to which this line can be shifted in both directions (see Figure 7). The two directions have different maximum shifts, and only one of them can be infinite.

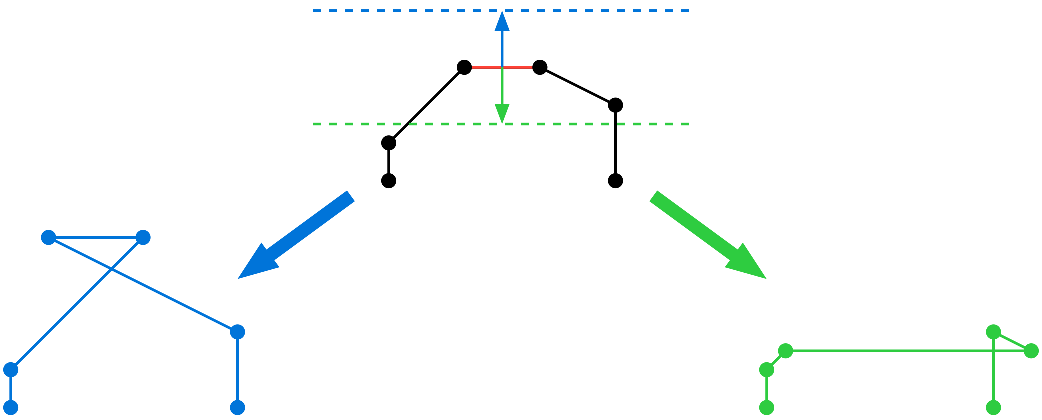
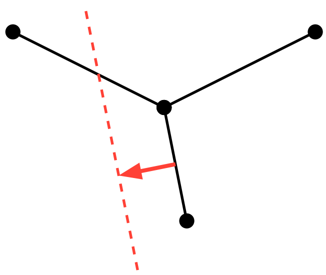
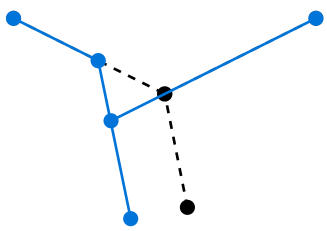


Figure 7: Illustration of an edge being shifted too far in the two directions.

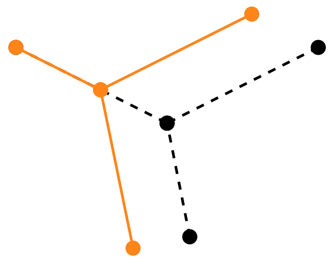
Additionally, since the polygons at hand represent buildings, several different polygons may share edges in situations where the buildings are adjacent. If that is the case in the initial configuration of polygons, our method also tries to keep these adjacencies. One way to consider it is to say that edges that are collinear should remain collinear. This is a rather limited constraint, as it still allows shared vertices to be split into two different vertices. The stronger version of this constraint is to enforce all shared vertices to remain the same vertex. This results in harder constraints that preserve the topology better at the cost of some freedom on the movements of the lines. In this case, if a vertex is shared by 3 non-collinear edges, keeping the vertex unique when shifting one of the edges implies to move at least one of the other edges, even if it was not necessary for the validity of the polygon. Figure 8 illustrates these different constraints on a simple example.



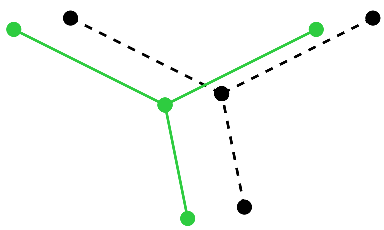
(a) Initial situation



(b) Softer constraint on shared edges.



(c) Harder constraint on shared vertices: a solution.



(d) Harder constraint on shared vertices: another solution.

Figure 8: Illustration of an edge being shifted with one of its vertex shared by three edges. Three different results are shown: Figure 8b displays the softer constraint on shared edges, while Figure 8c and Figure 8d show two different solutions to the harder constraint on shared vertices.

2.d. Optimisation and energy

Once a set of constraints are defined for the deformation of the polygons, this results in an infinite set of potential polygons. The goal is then to find the best polygon for a specific use case. This is called optimisation, and it usually needs two elements that will work together:

1. a criterion, also called energy, which is used to estimate how good a given polygon is,
2. an exploration strategy, which tries to navigate in the complex space of potential solutions to find the polygon giving the best energy.

In this report, the best energy is assumed to be the lowest energy, so the goal of optimisation is to find the polygon that minimises the energy. There are infinitely many ways to create a formula for an energy, and these different ways will result in a different polygon being the best. For the optimisation of the roofprints, the energy should be something that is minimal when the polygon corresponds perfectly to the edges of the roof. More details are given in *Chapter 3* about the energy used in our method.

Often, the energy can be a high-dimensional function, with many parameters to optimise together. In our case, with the constraints that we set on the polygons, there are usually n variables to optimise at the same time if the polygon has n vertices. Moreover, we consider that our starting point is already a good first guess: the initial outlines of our method are assumed to be up to a few meters off, with only small local deformations of the polygon being necessary. In this situation, the simplest procedure is to optimise every single parameter iteratively. In practice, this means:

1. picking one line,
2. computing the energy of the polygon obtained after shifting this line by many different amounts in both directions,
3. picking the polygon with the best energy as the new polygon,
4. starting again with a new line and the new polygon.

This kind of iterative and simplified process can however lead to completely incorrect final configurations. One of its downsides is that the first iterations are the most crucial: since it may start relatively far from its optimal solution, the first iterations may find their minimal values in the wrong direction, which can snowball into finding an incorrect locally minimal configuration. Moreover, the order of the iterations matter: the result may be completely different depending on which lines were treated first. This means that picking a good order is important, and also that it is possible to try different orders and pick the best final solution. In our case, treating the longest edges first seems to be a smart choice, as longer edges are relatively less impacted by their initial incorrect translation than smaller edges. But the main advantage of this method is its simplicity and its speed: it reduces the optimisation of a complex function into a series of simpler one-dimensional problems, which is computationally very quick.

2.e. 3D city modelling

Even if this thesis is focussed on producing 2D polygons, its outputs could greatly improve the generation of 3D building models. To understand why, it is necessary to know about the current existing methods to produce LoD2.2 buildings.

In the last few years, several methods have demonstrated their ability to produce high-quality LoD2.2 building models from two inputs: a high-density ALS point cloud and building outlines (Bauchet et al., 2024, Huang et al., 2022, Peđen et al., 2024). Since ALS data is acquired by planes shooting rays that are mostly vertical, it contains high density of points on horizontal surfaces (such as roofs) and low density on vertical surfaces (such as façades), as illustrated in Figure 9. This low density makes it difficult to use for the façades the same plane fitting techniques that are used for the roofs. Furthermore, the roof overhangs create occlusion which reduces further the amount of points on the façades. For these reasons, the methods mentioned above focus on creating a precise and accurate model of the roof and simply extrude down the roof to get the façades, which results in LoD2.2 models.

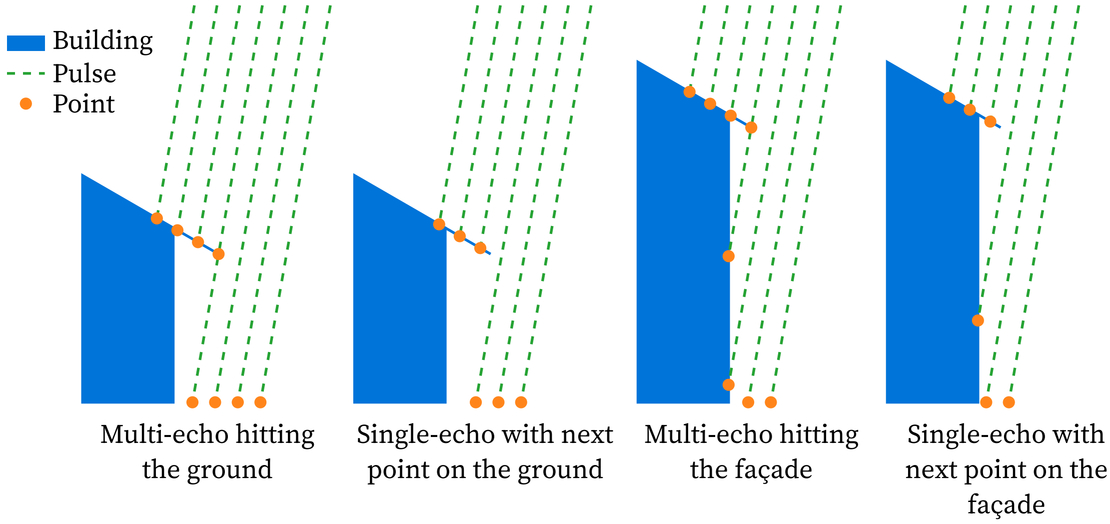


Figure 9: Illustration in profile view of the lack or sparsity of points on the façades.

In all of these methods, the outlines are used initially to select which points in the point cloud should be used for the reconstruction of each building. This is a crucial step, as having too many or too few points can lead to incorrect results. Therefore, to reconstruct a perfect roof, a roofprint would be the best input for these methods, even though the papers mention footprints due to the difference often not being made. On top of that, precise outlines could help extracting precise and well-oriented edges for the roof model, as extracting precise directions from ALS data is difficult.

Besides guiding the reconstruction process better with a roofprint, these methods could also use a footprint to create more detailed models. Instead of using the outline of the final 3D roof model to extrude down and get the final 3D building model, these methods could use the footprint and extrude it up until it reaches the roof model. This would allow to very simply move from LoD2.2 to LoD2.3, assuming that the footprints are correct.

3. Paper

This page is left intentionally blank, with the paper starting in the next page. The paper is fully self-contained, with its own introduction and conclusion, its own numbering and its own references.

4. Conclusion

How to generate coherent building roofprints and footprints from high-density ALS point clouds and existing imprecise outlines?

- How to identify and use the points on roof edges in ALS point clouds?
- How to identify and use the points in an ALS point cloud that contain information about the façades despite their sparsity?
- How to deform an imprecise outline with global and local transformations while preserving the angles of the edges?

To answer the main research question, I proposed a method to turn imprecise outlines into roofprints and footprints sequentially from ALS data. The same general idea is used for the two outlines: first identify the points containing information about the outline, and then iteratively deform the initial imprecise outline to fit with the selected points. The resulting roofprints were tested successfully on the French national datasets BD TOPO and LiDAR HD, while the footprints were only qualitatively assessed and still require further work.

Regarding points on roof edges, I showed how most of them can be identified with simple topology-aware operations on ALS point clouds. These points can then be used to compute a roofprint in 2D, without having to manipulate the points in 3D for the optimisation of the polygon. It is however necessary to compute for each point on the roof edge the orientation of the building it corresponds to, in order to prevent the points from a given building to be used by another building.

As for the points containing information about the façades, I showed that it becomes significantly easier to identify them once a 3D roof model is obtained using the roofprint computed previously. This allows to simply select all the points below the roof, a simple solution that works well despite the sparsity of the ALS data on the façades. Therefore, the order in which creating the roofprint and the footprint is of great importance. Being able to use at the same time the points on the ground and the points on the façades proved to be challenging but very promising, as the points hitting the ground under the roof are very valuable and sometimes the only available information about the façade.

The question of preserving the interesting properties of the input outlines was tackled by developing a robust deformation algorithm enforcing the necessary constraints. This algorithm is capable of preserving the angles of the edges and the validity of the polygons while still leaving a lot of freedom for deformations. This algorithm proved to be very effective to tackle at the same time globally translating the outlines and locally modifying the relations between neighbouring edges. By iteratively focussing on edges one by one, it also reduces a complex problem with 2 dimensions of freedom per point into a succession of 1-dimensional problems.

This thesis overall fits perfectly into the programme of the MSc Geomatics for the Built Environment. Buildings and their different representations are at the core of some courses. The combination of different data sources is also at the core of the programme, in this case high-density ALS point clouds, which are becoming increasingly available in several countries, with the existing building outlines datasets, which many countries have in one form or another. Developing this whole method took a lot of the knowledge acquired during the master, such as how lidar works and how it affects the structure and properties of ALS point clouds, how validity of polygons is crucial to many applications, and how current methods to reconstruct 3D models of buildings work.

References

This section only contains the references mentioned outside of Chapter 3. They are treated independently of the references in Chapter 3, meaning that there could be duplicates.

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A. Declaration of AI/LLM usage

Some AI tools were used during this thesis. The main usage comes from the use of GitHub Copilot, which was used to assist in writing code at all stages of the project. Then, Perplexity was used to look for explanation of concepts and to find relevant literature to start with at different stages of the project. No AI tool was used when writing this report.

All methods, results, analyses, and conclusions are my own, and I verified that the content is original and meets academic integrity standards.

B. Reproducibility

All data used during this thesis is openly available on the websites where they are distributed (data.gouv.fr):
- The BD TOPO dataset is available at data.gouv.fr/datasets/bd-topo-r
- The LiDAR HD dataset is available at data.gouv.fr/datasets/nuages-de-points-lidar-hd

An implementation of the method proposed in this thesis is available at github.com/alexandre-bry/MSc_Thesis-Code. All instructions to run the code and reproduce the results are available there, including the versions of the dependencies, and commands to download the data from the official sources. The validation dataset will also be published, and the link will be displayed in the same repository.

There is also another repository containing the notes, weekly reports and other materials produced during the thesis at github.com/alexandre-bry/MSc_Thesis-Report. The content of this repository is also available in the form of a website, which link can be found in the GitHub repository.

C. Glossary

Datasets

BAG (Basisregistratie Adressen en Gebouwen) — The Dutch dataset of buildings outlines with addresses and other attributes. More information at https://www.kadaster.nl/zakelijk/registraties/basisregistraties/bag. 1

3DBAG — An open 3D building data set that is generated fully automatically based on lidar data and covers the whole of the Netherlands. More information at https://docs.3dbag.nl/en/. 1, 2

AHN (Actueel Hoogtebestand Nederland) — A programme to produce high-density point clouds on the whole Dutch territory, with a new version every few years. More information at https://www.ahn.nl/. 1, 2, 4, 5

BD TOPO — The French dataset that contains (among many other types of objects) the buildings outlines, combining footprints and roofprints. More information at https://www.data.gouv.fr/datasets/bd-topo-r. i, 2, 12, A-2

LiDAR HD — The first project to collect high-density point clouds on the whole territory of France (except Guyane). More information at https://www.data.gouv.fr/datasets/nuages-de-points-lidar-hd. i, 1, 2, 4, 5, 6, 12, A-2

JUNN — A French public project officially launched in 2026 aiming at creating a nationwide digital twin. More information at https://junn-france.fr/. 1

Entities

TU Delft (Delft University of Technology) — The Dutch university in which this project was carried on. More information at https://www.tudelft.nl/en/. ii, 1, 3

IGN (Institut national de l'information géographique et forestière) — The reference public operator for geographic and forest information in France. More information at https://www.ign.fr/institut/identity-card. ii, 1, 2

Software

roofer — An algorithm to reconstruct a 3D building model from a point cloud and a 2D roofprint polygon in different LoDs up to 2.2. More information at https://github.com/3DBAG/roofer. 1, 2, 5

Vocabulary

footprint — The 2D outer boundary defined by the vertical projection of the outer walls/façades of a building. i, 2, 3, 4, 5, 10, 12, A-3

roofprint — The 2D outer boundary defined by the vertical projection of the roof of a building. i, 2, 3, 4, 5, 9, 10, 12, A-3

outline — Purposefully vague term designing either a roofprint or a footprint or when they are mixed and the differentiation cannot be made. i, 1, 2, 3, 4, 9, 10, 12, A-3

LoD (Level of Detail) — A concept to differentiate multi-scale representations of semantic 3D city models. It is in practice principally used to indicate the geometric detail of a model, primarily of buildings. More information at https://3d.bk.tudelft.nl/lod/. i, 1, 2, 4, 5, 9, 10, A-3

lidar (Light Detection and Ranging) — A method for determining ranges by targeting an object or a surface with a laser and measuring the time for the reflected light to return to the receiver. (Wikipedia) More information at https://en.wikipedia.org/wiki/Lidar. 5, 12, A-3, A-4

ALS (Airborne Laser Scanning) — Aircraft-mounted lidar that emits laser pulses toward the ground, capturing millions of georeferenced points for large-scale terrain and vegetation models. More information at https://en.wikipedia.org/wiki/Lidar. i, 1, 3, 4, 5, 6, 9, 10, 12

MLS (Mobile Laser Scanning) — Vehicle-mounted lidar that records dense point clouds while moving, used for detailed street-level 3-D mapping. More information at https://en.wikipedia.org/wiki/Lidar. 4

TLS (Terrestrial Laser Scanning) — Tripod-mounted static lidar that scans surroundings from fixed positions, ideal for high-resolution models of structures and heritage sites. More information at https://en.wikipedia.org/wiki/Lidar. 4